Maximal overgroups of a full negative $6+2$ torus element

L045 recovery: a classification interface

Let $q = p^f \geq 4$, $d = (2, q-1)$, $A = q^3+1$, and $B = q+1$. Let $V = V_6^- \perp V_2^-$ over $\mathbb{F}_q$, and let $X = g \perp h$ where g and h generate the norm-one tori of orders A and B on the two summands. Thus $X \in \Omega_8^+(q)$ and its image $x \in G = P\Omega_8^+(q)$ has order A/d, by the acquired torus proof. The primitive elements g, h are arbitrary; no relation between their choices is assumed.

Theorem 1. *Every maximal subgroup of G containing x is one of the following:*

1. *the unique stabilizer D of $V_6^- \perp V_2^-$;*
2. *the normalizer of a unitary structure on a four-dimensional $\mathbb{F}_{q^2}$-space; there are at most two such subgroups containing x;*
3. *a subgroup of type B_3: $\Omega_7(q)$ for odd q, or $\mathrm{Sp}_6(q)$ for even q, in one of the three triality actions.*

The natural-action B_3 subgroups do not contain x. Also $N_G(\langle x \rangle) \leq D$, so a separate cyclic-normalizer term is not missing from this cover.

The only classification premise is the complete published maximal-subgroup classification in Bray–Holt–Roney-Dougal (BHRD), Main Theorem 2.1.1, Section 8.1 and Table 8.50 [1]. The table is explicitly attributed there to Kleidman's classification of these orthogonal groups. We use the table for the simple group G, not for arbitrary almost simple extensions. Its novelty rows therefore give no additional maximal subgroup of G. All eliminations for our particular elements are proved below. The complete row-by-row routing is retained separately.

Lemma 2. *There is a prime $r \mid |x|$ with $\mathrm{ord}_r(p) = 6f$. Every such prime satisfies $r \geq 6f+1$ and $\mathrm{ord}_r(q) = 6$. Moreover the entire primitive part $\Phi_{6f}^*(p)$ divides $|x|$.*

Proof. The primitive-divisor theorem applied to $p^{6f}-1$ has no exception here: the exceptional pair $(p, 6f) = (2, 6)$ would mean $q = 2$. All primitive prime powers divide $p^{3f}+1 = A$, and none divides 2. This proves the assertions after passing to the projective quotient of order $d \leq 2$. We retain the classical primitive-divisor theorem as a named number-theoretic input. Its exact statement and the primitive-part convention are also recalled in the acquired Bamberg–Penttila preprint [2, pp.2–3]. □

Proof of the type assertion in Theorem 1. Apply the complete BHRD table to the preimage of a maximal subgroup in $\Omega_8^+(q)$. Central factors of order at most two do not affect the primitive divisor. The parabolic rows have Levi factors of types $\mathrm{GL}_4(q)$, $\mathrm{GL}_2(q)\Omega_4^+(q)$ or $\mathrm{GL}_3(q)$; their orders have no factor r. The same test excludes the positive $2+6$ and SL_4 rows, the $3+5$ and $\mathrm{Sp}_2 \circ \mathrm{Sp}_4$ rows, the four-plane decompositions, the two four-space decompositions, and $\Omega_4^+(q^2)$. Their non-characteristic order factors divide products of q^j-1 for $j \leq 4$, together with powers of 2 and 3. The additional dihedral novelty row has factors q^2+1 and powers of two, and is excluded by the same argument. The negative $2+6$ and unitary four-space rows survive.

The subfield rows are excluded using the *absolute* order $6f$. For $q = q_0^s$, $q_0 = p^a$, $s \geq 2$, the factors in $\Omega_8^+(q_0)$ have exponents at most $6a < 6f$. For the two $\Omega_8^-(q_0)$ actions with $q = q_0^2$, the exponents are at most $8a < 12a = 6f$. For ${}^3D_4(q_0)$ with $q = q_0^3$, use

$$|{}^3D_4(q_0)| = q_0^{12}(q_0^8+q_0^4+1)(q_0^6-1)(q_0^2-1), \qquad q_0^8+q_0^4+1 \mid q_0^{12}-1.$$

Here $12a < 18a = 6f$. All displayed extension factors in these table rows have order a power of two. The $L_3(q).3$ row has factors of exponents at most three and an extra factor three, so is excluded too.

Consider the $U_3(q).3$ row, which occurs only when $q \equiv 2 \pmod 3$. In its projective group, $y = x^3$ lies in $\mathrm{PSU}_3(q)$ and has order $A/(3d)$. An r-element in $\mathrm{PSU}_3(q)$ acts irreducibly on its natural three-space over $\mathbb{F}_{q^2}$, since $\mathrm{ord}_r(q^2) = 3$. Its centralizer is the norm-one Singer torus modulo scalars, of order $(q^2-q+1)/3$. This follows directly by identifying the commuting algebra with $\mathbb{F}_{q^6}$: the unitary condition is norm one to $\mathbb{F}_{q^3}$, the determinant-one condition leaves order q^2-q+1, and the scalar center has order three. Projective centralization gives no extra elements: lift the r-element with order r; a commuting lift can differ by a scalar of order three, which its rth power forces to be one. Since y centralizes its own r-subgroup, its order must divide this torus order. This is impossible, their ratio being $B/d > 1$.

The remaining bounded-order rows are handled as follows. The monomial and extraspecial rows have a normal 2-group with quotient contained in S_8 (or $L_3(2)$), and require $q = p$ odd. Their primitive part is therefore at most seven: a primitive divisor is at least seven, and seven occurs only once in those quotient orders. The same assertion holds for the row $2 \cdot \Omega_8^+(2)$, whose prime divisors are $2, 3, 5, 7$, with only one factor seven. But

$$\Phi_6^*(p) = \frac{p^2 - p + 1}{(3, p+1)} > 7 \qquad (p \geq 7).$$

Only $q = 5$ remains in our range. Now $|x| = 63$. An odd-order element injects through a 2-group kernel; S_8 and $L_3(2)$ have no element of order 63. Neither does $\Omega_8^+(2)$, as proved in Lemma 3 below. The exceptional rows $2 \cdot \mathrm{Sz}(8)$ and the two central extensions of A_{10} occur only at $q = 5$. The first group has order prime to three; the second ones cannot have order 63, since a permutation of that order requires disjoint cycles of lengths nine and seven. The A_9 row has $q = 2$ and is outside the hypothesis. The G_2 row is marked as a novelty and is not maximal in the simple group G. Thus only the three stated types remain. A natural B_3 subgroup fixes a nonsingular line, whereas X has no eigenvalue in $\mathbb{F}_q$. □

Lemma 3. *No element of the orthogonal group $O_8^+(2)$ has order* 63.

Proof. Such an element would be semisimple. An eigenvalue whose order is divisible by 63 has degree at least six, and an eigenvalue of order 63 has degree six. Its irreducible polynomial is not self-reciprocal: no power of 2 is -1 modulo 63. A second, distinct dual constituent would require twelve dimensions. If no eigenvalue has order divisible by 63, separate constituents must supply the factors nine and seven. Order nine requires six dimensions, since $\mathrm{ord}_9(2) = 6$. A constituent supplying the factor seven costs at least six dimensions: if its degree is three, its eigenvalue has order seven and a distinct dual constituent is required; otherwise its degree is at least six (this includes eigenvalue order 21). The two contributions are distinct, since otherwise an eigenvalue would have order divisible by 63. Again at least twelve dimensions are necessary. □

Proof of the multiplicities and normalizer assertion. Both X and X^2 have exactly two distinct irreducible constituents, of dimensions six and two. Indeed g^2 has order $A/(2, A)$, which exceeds $q^2 - 1$ and does not divide $q^3 - 1$; its degree is therefore six. The order of h^2 is $B/(2, B) > 2$ for $q \geq 4$, so its degree is two. Thus the only proper nonzero X-invariant subspaces are V_6 and V_2, and the decomposition stabilizer D is unique.

Let $E \cong \mathbb{F}_{q^2}$ be a quadratic scalar-field structure whose unitary normalizer contains X. Conjugation by X induces an automorphism of E of order at most two, so X^2 commutes with E. Consequently

$$E \subseteq C_{\mathrm{End}_{\mathbb{F}_q}(V)}(X^2) = \mathbb{F}_{q^6} \oplus \mathbb{F}_{q^2}.$$

This algebra is commutative and contains X, so X in fact centralizes E. Each coordinate projection of E is an embedding. Up to a common Frobenius automorphism of E, there are exactly

two choices, determined by the relative Frobenius on the two factors. A field structure and the quadratic form determine its full unitary extension normalizer; hence at most two of the unitary maximal subgroups can contain x.

Finally a lift of an element normalizing $\langle x\rangle$ conjugates X to ϵX^j, where $\epsilon \in \{1,-1\}$ and $(j, A/d) = 1$. The two constituents of X^j still have degrees six and two: $(j, A) \le d$, so their orders lose at most a factor two, and multiplication by ϵ does not change their degrees. The preceding degree argument applies. Their unique dimensions force the lift to preserve both summands. Therefore $N_G(\langle x\rangle) \le D$. □

Scope and attribution. This is an application of an explicitly retained complete subgroup classification, not a new classification. It corrects the missing arbitrary-element interface in the use of Leemans–Liebeck Table 4; it does not infer that that source's multiplicities or projective torus orders are correct. The triality analysis deciding which B_3 subgroups contain a particular x, the inverter bounds and the final generating-pair inequality are separate components. No whole-original admission, measurement or new question is asserted here. The torus construction and the surrounding orthogonal program were authored by the root agent; this cover specialization and its two-field-structure argument were authored by publication_recovery. The parent independently obtained the relative-field orientation argument in a separate orthogonal manuscript. The peer f003_resume supplied a supporting elementary order-21/63 note and arithmetic checks; these are retained as a contribution, not as an independent review of this component.

References

[1] J. N. Bray, D. F. Holt and C. M. Roney-Dougal, *The Maximal Subgroups of the Low-Dimensional Finite Classical Groups*, LMS Lecture Note Series 407, Cambridge University Press, 2013. Main Theorem 2.1.1, Section 8.1, Table 8.50 (printed pp.55,373–374,402–404). Complete published book acquired from the National Academic Digital Library; the classification is retained, not recomputed.

[2] J. Bamberg and T. Penttila, *Overgroups of cyclic Sylow subgroups of linear groups*, Comm. Algebra 36 (2008), 2503–2543; complete 27-page author-hosted preprint acquired from Bamberg's Ghent preprint page, with its old-draft qualification and current author errata explicitly retained. Only its elementary primitive-part convention is used above; the complete maximal cover comes from BHRD.